



Prepared By:

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1. Prime Number Checking

```
#include<stdio.h>
int main()
{
    int n,i,c=0;
    printf("Enter the number to be checked:");
    scanf("%d",&n);
    for(i=2;i<=(n/2);i++)
    {
        if(n%i==0)
        {
            c++;
            break;
        }
    }
    if(c==1)
    {
        printf("Not Prime");
    }
    else
    {
        printf("Prime");
    }
    return 0;
}
```

2. Prime Number List in range

```
#include<stdio.h>
int main()
{
    int n,m,i,j,c;
    printf("Enter the lower range:");
    scanf("%d",&n);
    printf("Enter the Upper range:");
    scanf("%d",&m);

    for(i=n;i<=m;i++)
    {
        c=0;
        for(j=2;j<=(i/2);j++)
        {
            if(i%j==0)
            {
                c++;
                break;
            }
        }
        if(c==0)
        {
            printf("%d\t",i);
        }
    }
    return 0;
}
```

3. Prime Number Checking using recursion

```
#include<stdio.h>

int IsPrime(int n, int i);

int main()
{
    int n,i,c;
    printf("Enter the number to be checked:");
    scanf("%d",&n);
    c=IsPrime(n,n/2);
    if(c==0)
    {
        printf("Not Prime");
    }
    else
    {
        printf("Prime");
    }
}

int IsPrime(int n, int i)
{
    if(i==1)
    {
        return 1;
    }
    else
    {
        if(n%i==0)
            return 0;
        else
            IsPrime(n,i-1);
    }
}
```

4. Factorial Using Recursion

```
#include<stdio.h>

int Factorial(int n);

int main()
{
    int n,m;
    printf("Please Enter the Number: ");
    scanf("%d",&n);
    m=Factorial(n);
    printf("The Factorial of %d is %d", n, m);
    return 0;
}

int Factorial(int n)
{
    if(n<=1)
        return 1;
    else
        return n*Factorial(n-1);
}
```

5. Fibonacci Series

```
#include<stdio.h>
int main()
{
    int a=0,b=1,temp,i,n;
    printf("Enter the No of Terms: ");
    scanf("%d", &n);
    printf("The series is %d\t%d\t",a,b);
    temp=a+b;
    for(i=3;i<=n;i++)
    {
        printf("%d\t",temp);
        a=b;
        b=temp;
        temp=a+b;
    }
    return 0;
}
```

6. Fibonacci series using Recursion

```
#include<stdio.h>
int fibonacci(int n);

int main()
{
    int n,i;
    printf("Enter the No of Terms: ");
    scanf("%d",&n);
    printf("The series is : ");
    for(i=0;i<n;i++)
    {
        printf("%d\t",fibonacci(i));
    }
    return 0;
}

int fibonacci(int n)
{
    if(n<=1)
        return n;
    else
        return fibonacci(n-1)+fibonacci(n-2);
}
```

7. Factors of a Number

```
#include<stdio.h>
int main()
{
    int n,i;
    printf("Enter The Number: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            printf("%d\t",i);
        }
    }
    return 0;
}
```

8. Factors of Number Using Recursion

```
#include<stdio.h>
void factor(int n, int i);

int main()
{
    int n,i;
    printf("Enter The Number: ");
    scanf("%d",&n);
    factor(n,1);
    return 0;
}

void factor(int n,int i)
{
    if(i<=n)
    {
        if(n%i==0)
        {
            printf("%d\t",i);
        }
        factor(n,i+1);
    }
}
```

9. Palindrome Number Checking

```
#include<stdio.h>
int main()
{
    int n,a,temp=0,i;
    printf("Enter the number");
    scanf("%d",&n);
    a=n;
    while(n!=0)
    {
        temp=(temp+(n%10));
        n=n/10;
        if(n==0)
        {
            break;
        }
        else
        {
            temp=temp*10;
        }
    }
    if(a==temp)
    {
        printf("Palindrome");
    }
    else
    {
        printf("Not palindrome");
    }
    return 0;
}
```

10. Armstrong Number

```
#include<stdio.h>
int main()
{
    int n,c=0,a,b,sum=0;
    printf("Enter the Number: ");
    scanf("%d",&n);
    a=n;
    b=n;
    while(n!=0)
    {
        c++;
        n=n/10;
    }
    while(b!=0)
    {
        sum=sum+pow((b%10),c);
        b=b/10;
    }
    if(a==sum)
    {
        printf("Armstrong Number");
    }
    else
    {
        printf("Not Armstrong Number");
    }
    return 0;
}
```

11.Swap without third variable

```
#include<stdio.h>
int main()
{
    int x,y;
    x=5;
    y=10;

    x=x+y;
    y=x-y;
    x=x-y;

    printf("x=%d  y=%d",x,y);
    return 0;
}
```

12. Perfect Number Checking

```
#include<stdio.h>
int main()
{
    int n,i,sum=0;
    printf("Enter the Number: ");
    scanf("%d",&n);
    for(i=1;i<n;i++)
    {
        if(n%i==0)
        {
            sum=sum+i;
        }
    }
    if(n==sum)
    {
        printf("Perfect Number");
    }
    else
    {
        printf("Not Perfect Number");
    }
    return 0;
}
```

13. Perfect Number Checking Using Recursion

```
#include<stdio.h>
int perfect(int n, int i);

int main()
{
    int n,m;
    printf("Enter the Number: ");
    scanf("%d", &n);
    m=perfect(n,n/2);
    if(m==n)
    {
        printf("Perfect Number");
    }
    else
    {
        printf("Not Perfect Number");
    }
    return 0;
}

int perfect(int n, int i)
{
    if(i==1)
        return 1;
    else
    {
        if(n%i==0)
            return i+perfect(n,i-1);
        else
            return perfect(n,i-1);
    }
}
```

14.Sum of N terms of Natural Numbers using recursion

```
#include<stdio.h>
int Sum(int n);

int main()
{
    int n;
    printf("Enter the value of last number: ");
    scanf("%d",&n);
    printf("The Sum is %d",Sum(n));
    return 0;
}

int Sum(int n)
{
    if(n<=1)
        return 1;
    else
        return n+Sum(n-1);
}
```

15.Linear Search

```
#include<stdio.h>
int main()
{
    int a[20],i,n,m,c=0;
    printf("Enter the no of elements in the array:");
    scanf("%d",&n);
    printf("Enter the Array elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the No to be Searched");
    scanf("%d",&m);
    for(i=0;i<n;i++)
    {
        if(a[i]==m)
        {
            c=i;
            break;
        }
    }
    if(c!=0)
    {
        printf("Element Found in the  %dth index",c);
    }
    else
    {
        printf("Element Not Found");
    }
    return 0;
}
```

16.Array Sorting

```
#include<stdio.h>
int main()
{
    int a[20],i,j,n,temp;
    printf("Enter the no of elements in the array: ");
    scanf("%d",&n);
    printf("Enter the array Elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(a[j]<a[i])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    printf("The Sorted array is: ");
    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
    }
    return 0;
}
```

17. $(1 + 1/2 + 1/3 + 1/4 + \dots + 1/N)$

```
#include<stdio.h>
int main()
{
    float n,i,sum=0;
    printf("Enter The limit of the series: ");
    scanf("%f",&n);
    for(i=1;i<=n;i++)
    {
        sum=sum+(1/i);
    }
    printf("The sum is %.2f",sum);
    return 0;
}
```

18. $(1 + 1/2 + 1/3 + 1/4 + \dots + 1/N)$ Using Recursion

```
#include<stdio.h>
float sum(float n);

int main()
{
    float n,i,s;
    printf("Enter The limit of the series: ");
    scanf("%f",&n);
    s=sum(n);
    printf("The sum is %.2f",s);
    return 0;
}

float sum(float n)
{
    if(n==1)
        return 1;
    else if(n<=0)
        return 0;
    else
        return (1/n)+sum(n-1);
}
```

19. $1*3+2*5+3*7+.....+n*(2n+1)$

```
#include<stdio.h>
int main()
{
    int n,i,sum=0;
    printf("Enter the Value of N: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        sum=sum+(i*((2*i)+1));
    }
    printf("The sum is %d ", sum);
    return 0;
}
```

$$20.1 + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n)$$

```
#include<stdio.h>
int main()
{
    int n, i,j,sum=0;
    printf("Enter The limit of the series: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        for(j=1;j<=i;j++)
        {
            sum=sum+j;
        }
    }
    printf("The sum is %d ", sum);
    return 0;
}
```

21. Equation Solving and find roots

```
#include<stdio.h>
int main()
{
    float a,b,c,d, x1,x2,i1,i2;
    printf("Enter the value of a,b,c : ");
    scanf("%f %f %f",&a,&b,&c);

    d=(b*b)-(4*a*c);
    if(d>0)
    {
        x1=(-b+sqrt(d))/(2*a);
        x2=(-b-sqrt(d))/(2*a);
        printf("The first root is %.2f\n",x1);
        printf("The Second root is %.2f",x2);
    }
    else if(d==0)
    {
        x1=(-b/(2*a));
        printf("The first root is %.2f\n",x1);
        printf("The Second root is %.2f",x1);
    }
    else
    {
        x1=x2=(-b/(2*a));
        i1=i2=(sqrt(-d))/(2*a);
        printf("The first root is %.2f+i%.2f\n",x1,i1);
        printf("The Second root is %.2f - i%.2f\n",x2,i2);
    }
    return 0;
}
```

22. Permutation and Combination

```
#include<stdio.h>
int main()
{
    int a,b,c=1,d=1,e=1,i,p,m;
    printf("Enter the value of n and r:");
    scanf("%d %d",&a,&b);
    if(a>=b)
    {
        for(i=1;i<=a;i++)
        {
            c=c*i;
        }
        for(i=1;i<=(a-b);i++)
        {
            d=d*i;
        }
        for(i=1;i<=b;i++)
        {
            e=e*i;
        }
        p=c/d;
        m=c/(d*e);
        printf("The Permutation is: %d\n",p);
        printf("The Combination is: %d",m);
    }
    else
    {
        printf("Enter the correct value of n and r");
    }
    return 0;
}
```

23. Find the number of reoccurring of a given digit from a Number

```
#include<stdio.h>
int main()
{
    int a,b,c=0;
    printf("Enter The number: ");
    scanf("%d",&a);
    printf("Enter the reoccurring digit: ");
    scanf("%d",&b);
    while(a!=0)
    {
        if((a%10)==b)
        {
            c++;
        }
        a=a/10;
    }
    printf("%d times found",c);
    return 0;
}
```

24. Perfect Square Checking

```
#include<stdio.h>
int main()
{
    float a,b,i,c=0;
    printf("Enter the Number: ");
    scanf("%f",&a);
    b=sqrt(a);
    for(i=1;i<a;i++)
    {
        if(i==b)
        {
            c++;
            break;
        }
    }
    if(c==1)
    {
        printf("Perfect Square Number");
    }
    else
    {
        printf("Not Perfect Square");
    }
    return 0;
}
```

25.GCD & LCM

```
#include<stdio.h>
int main()
{
    int a,b,i,gcd,lcm;
    printf("Enter the value of A and B: ");
    scanf("%d %d",&a,&b);
    for(i=1;i<=a && i<=b;i++)
    {
        if(a%i==0 && b%i==0)
        {
            gcd=i;
        }
    }
    lcm=(a*b)/gcd;
    printf("The value of GCD is %d\n",gcd);
    printf("The value of lcm is %d",lcm);
    return 0;
}
```

26.GCD & LCM Using Recursion

```
#include<stdio.h>
int gcdlcm(int a,int b);
int main()
{
    int a,b,i,gcd,lcm;
    printf("Enter the value of A and B: ");
    scanf("%d %d",&a,&b);
    gcd=gcdlcm(a,b);
    lcm=(a*b)/gcd;
    printf("The value of GCD is %d\n",gcd);
    printf("The value of lcm is %d",lcm);
    return 0;
}
int gcdlcm(int a,int b)
{
    if(b!=0)
        return gcdlcm(b, a%b);
    else
        return a;
}
```

27. Permutation & Combination Using Recursion

```
#include<stdio.h>
int factorial(int n);
int main()
{
    int a,b,c=1,d=1,e=1,i,p,m;
    printf("Enter the value of n and r:");
    scanf("%d %d",&a,&b);
    if(a>b)
    {
        c=factorial(a);
        d=factorial(a-b);
        e=factorial(b);
        p=c/d;
        m=c/(d*e);
        printf("The Permutation is: %d\n",p);
        printf("The Combination is: %d",m);
    }
    else
    {
        printf("Enter the correct value of n and r");
    }
    return 0;
}

int factorial(int n)
{
    if(n<=1)
        return 1;
    else
        return n*factorial(n-1);
}
```

28. Insert an element to array

```
#include<stdio.h>
int main()
{
    int a[20],n,i,m,p;
    printf("Enter how many numbers in the array : ");
    scanf("%d",&n);
    printf("Enter the array elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the position to be insert: ");
    scanf("%d",&p);
    printf("Enter the value to be inserted");
    scanf("%d",&m);
    for(i=n-1; i>=(p-1);i--)
    {
        a[i+1]=a[i];
    }
    a[p-1]=m;
    for(i=0;i<=n;i++)
    {
        printf("%d\t",a[i]);
    }
    return 0;
}
```

29.Remove an element from Array

```
#include<stdio.h>
int main()
{
    int a[20],n,i,pos;
    printf("Enter the No of elements in the array: ");
    scanf("%d",&n);
    printf("Enter the Array elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the Position to deleted: ");
    scanf("%d",&pos);
    if(pos>=n+1)
    {
        printf("Deletion is not possible");
    }
    else
    {
        for(i=pos-1;i<n-1;i++)
        {
            a[i]=a[i+1];
        }
        printf("The Resultant Array is: ");
        for(i=0;i<n-1;i++)
        {
            printf("%d\t",a[i]);
        }
    }
    return 0;
}
```

30.Remove Duplicates From an Array

```
#include<stdio.h>
int main()
{
    int a[20],i,n,j,k;
    printf("Enter the number of elements in the array: ");
    scanf("%d",&n);
    printf("Enter the array Elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(a[i]==a[j])
            {
                for(k=j;k<n;k++)
                {
                    a[k]=a[k+1];
                }
                n--;
                j--;
            }
        }
    }
    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
    }
    return 0;
}
```

31.Decimal to Binary

```
#include<stdio.h>
int main()
{
    int n,temp=0,i=1;
    printf("Enter the decimal number: ");
    scanf("%d",&n);
    while(n!=0)
    {
        temp=temp+((n%2)*i);
        i=i*10;
        n=n/2;
    }
    printf("The Binary value is: %d",temp);
    return 0;
}
```

32.Binary to Decimal

```
#include<stdio.h>
int main()
{
    int a,temp=0,i=0;
    printf("Enter the Binary Number");
    scanf("%d",&a);
    while(a!=0)
    {
        temp=temp+(a%10)*pow(2,i);
        i++;
        a=a/10;
    }
    printf("The decimal value is %d",temp);
    return 0;
}
```

33.Decimal to Octal

```
#include<stdio.h>
int main()
{
    int n,temp=0,i=1;
    printf("Enter the decimal number: ");
    scanf("%d",&n);
    while(n!=0)
    {
        temp=temp+((n%8)*i);
        i=i*10;
        n=n/8;
    }
    printf("The octal value is: %d",temp);
    return 0;
}
```

34.Octal to decimal

```
#include<stdio.h>
int main()
{
    int a,temp=0,i=0;
    printf("Enter the octal Number : ");
    scanf("%d",&a);
    while(a!=0)
    {
        temp=temp+(a%10)*pow(8,i);
        i++;
        a=a/10;
    }
    printf("The decimal value is %d",temp);
    return 0;
}
```

35.Decimal to Hexadecimal

```
#include<stdio.h>
int main()
{
    int n,temp,i=1,j;
    char hexa[100];
    printf("Enter the decimal number: ");
    scanf("%d",&n);
    while(n!=0)
    {
        temp=n%16;
        if(temp<10)
            temp=temp+48;
        else
            temp=temp+55;
        hexa[i++]=temp;
        n=n/16;
    }
    for(j=i-1;j>0;j--)
    {
        printf("%c",hexa[j]);
    }
    return 0;
}
```

36.Hexadecimal to Decimal

```
#include<stdio.h>
int main()
{
    char hexa[20];
    int i, val, len, decimal=0;
    printf("Enter the Hexadecimal Value: ");
    gets(hexa);
    len=strlen(hexa);
    len--;
    for(i=0;hexa[i]!='\0';i++)
    {
        if(hexa[i]>='0' && hexa[i]<='9')
        {
            val=hexa[i]-48;
        }
        else if(hexa[i]>='a' && hexa[i]<='f')
        {
            val=hexa[i]-87;
        }
        else if(hexa[i]>='A' && hexa[i]<='F')
        {
            val=hexa[i]-55;
        }
        decimal=decimal+val*pow(16,len);
        len--;
    }
    printf("The decimal value is %d",decimal);
    return 0;
}
```

37.Triangle -01

```
*  
* *  
* * *  
* * * *  
* * * * *
```

```
#include<stdio.h>  
int main()  
{  
    int i,j,n;  
    printf("Enter the row number: ");  
    scanf("%d",&n);  
    for(i=0;i<n;i++)  
    {  
        printf("\n");  
        for(j=0;j<=i;j++)  
        {  
            printf("*");  
        }  
    }  
    return 0;  
}
```

38.Triangle-02

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
#include<stdio.h>
int main()
{
    int i,j,n;
    printf("Enter the row number: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("\n");
        for(j=1;j<=i;j++)
        {
            printf("%d",j);
        }
    }
    return 0;
}
```

39.Triangle-03

```
* * * * *
* * * *
* * *
* *
*
```

```
#include<stdio.h>
int main()
{
    int n,i,j;
    printf("Enter the row Number: ");
    scanf("%d",&n);
    for(i=n;i>=1;i--)
    {
        printf("\n");
        for(j=i;j>=1;j--)
        {
            printf("*");
        }
    }
    return 0;
}
```

40.Triangle-04

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
#include<stdio.h>
int main()
{
    int n,i,j;
    printf("Enter the row Number: ");
    scanf("%d",&n);
    for(i=n;i>=1;i--)
    {
        printf("\n");
        for(j=1;j<=i;j++)
        {
            printf("%d",j);
        }
    }
    return 0;
}
```

41.Floyds Triangle

```
1
2 3
4 5 6
7 8 9 10
```

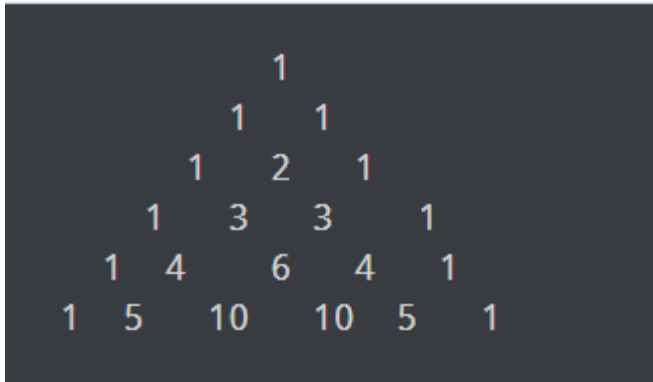
```
#include<stdio.h>
int main()
{
    int n,i,j,number=1;
    printf("Enter the Number of row: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("\n");
        for(j=1;j<=i;j++)
        {
            printf("%d\t",number);
            number++;
        }
    }
    return 0;
}
```

42. Pyramid of star



```
#include<stdio.h>
int main()
{
    int n,i,j;
    printf("Enter the no of rows: ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("\n");
        for(j=1;j<=((2*n)-1);j++)
        {
            if(j>=(n-(i-1)) && j<=(n+(i-1)))
                printf("*");
            else
                printf(" ");
        }
    }
    return 0;
}
```

43.Pascal triangle



```
#include <stdio.h>
int main() {
    int rows, coef = 1, space, i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &rows);
    for (i = 0; i < rows; i++) {
        for (space = 1; space <= rows - i; space++)
            printf(" ");
        for (j = 0; j <= i; j++) {
            if (j == 0 || i == 0)
                coef = 1;
            else
                coef = coef * (i - j + 1) / j;
            printf("%4d", coef);
        }
        printf("\n");
    }
    return 0;
}
```

44. Multiplication Table

```
#include<stdio.h>
int main()
{
    int n,i,s;
    printf("Enter the number: ");
    scanf("%d",&n);
    for(i=1;i<=10;i++)
    {
        printf("%d x %d = %d \n",n,i,(n*i));
    }
    return 0;
}
```

45.Strong number checking

```
#include<stdio.h>
int factorial(int n);

int main()
{
    int n, temp=0, b;
    printf("Enter the Number: ");
    scanf("%d",&n);
    b=n;
    while(n!=0)
    {
        temp=temp+factorial(n%10);
        n=n/10;
    }
    if(b==temp)
    {
        printf("Strong Number");
    }
    else
    {
        printf("Not Strong Number");
    }
    return 0;
}

int factorial(int n)
{
    if(n<=1)
        return 1;
    else
        return n*factorial(n-1);
}
```

46.Second Maximum Element in Array

```
#include<stdio.h>
int main()
{
    int a[20],i,j,n,temp;
    printf("Enter the no of elements in the array: ");
    scanf("%d",&n);
    printf("Enter the array Elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(a[j]>a[i])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    printf("The Second Maximum Element is: ");
    printf("%d\t",a[1]);
    return 0;
}
```

47. In an array, Summation of pair = given value

```
#include<stdio.h>
int main()
{
    int a[20],n,i,j,c=0,m;
    printf("Enter the number of elements in the array: ");
    scanf("%d",&n);
    printf("Enter the elements of the array : ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the compared value: ");
    scanf("%d",&m);
    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if((a[i]+a[j])==m)
            {
                printf("{%d , %d}",a[i],a[j]);
            }
        }
    }
    return 0;
}
```

48.String Length without Library Function

```
#include<stdio.h>
int main()
{
    char a[100];
    int c=0;
    printf("Enter the String : ");
    gets(a);
    while(a[c]!='\0')
    {
        c++;
    }
    printf("The Length is %d",c);
    return 0;
}
```

49.String Reverse without Library function

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int l=0,i,j;
    printf("Enter the string : ");
    gets(a);
    while(a[l]!='\0')
    {
        l++;
    }
    for(i=0,j=l-1;i<l,j>=0;i++,j--)
    {
        b[j]=a[i];
    }
    for(j=0;j<l;j++)
    {
        printf("%c",b[j]);
    }
    return 0;
}
```

50.String Palindrome

```
#include<stdio.h>
#include<string.h>
int main()
{
    int a[100],b[100];
    int i,j,n,c=0;
    printf("Enter the string: ");
    gets(a);
    strcpy(b,a);
    strrev(b);
    if(strcmp(a,b)==0)
    {
        printf("palindrome");
    }
    else
    {
        printf("Not Palindrome");
    }
    return 0;
}
```

51.String Copy without Library Function

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int n=0,i;
    gets(a);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0;i<n;i++)
    {
        b[i]=a[i];
    }
    for(i=0;i<n;i++)
    {
        printf("%c",b[i]);
    }
    return 0;
}
```

52.String Palindrome without any library function

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int n=0,i,j,c=0;
    gets(a);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0,j=n-1;i<n,j>=0;i++,j--)
    {
        b[j]=a[i];
    }
    for(i=0;i<n;i++)
    {
        if(a[i]==b[i])
        {
            c++;
        }
    }
    if(c==n)
    {
        printf("Palindrome");
    }
    else
    {
        printf("Not Palindrome");
    }
    return 0;
}
```

53. String Sorting

```
#include<stdio.h>
int main()
{
    char a[100],temp;
    int n=0,i,j;
    gets(a);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0;i<n;i++)
    {
        for(j=0;j<=i;j++)
        {
            if(a[j]>a[i])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    for(i=0;i<n;i++)
    {
        printf("%c",a[i]);
    }
    return 0;
}
```

54. Anagram Checking

```
#include<stdio.h>
int main()
{
    char a[100],b[100],temp;
    int n=0,m=0,i,j,c=0;
    gets(a);
    gets(b);
    while(a[n]!='\0')
    {
        n++;
    }
    while(b[m]!='\0')
    {
        m++;
    }
    if(n!=m)
    {
        printf("Not Anagram");
    }
    else
    {
        for(i=0;i<n;i++)
        {
            for(j=0;j<=i;j++)
            {
                if(a[j]>a[i])
                {
                    temp=a[i];
                    a[i]=a[j];
                    a[j]=temp;
                }
            }
        }
        for(i=0;i<m;i++)
        {
            for(j=0;j<=i;j++)
            {
                if(b[j]>b[i])
```

```

        {
            temp=b[i];
            b[i]=b[j];
            b[j]=temp;
        }
    }
}
for(i=0;i<n;i++)
{
    if(a[i]==b[i])
    {
        c++;
    }
}
if(c==m)
{
    printf("Anagram");
}
else
{
    printf("Not Anagram");
}
}
return 0;
}

```

55.Count Word from a string

```
#include<stdio.h>
int main()
{
    char a[100];
    int n=0,i,c=0;
    gets(a);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0;i<n;i++)
    {
        if(a[i]==' ' && a[i+1]!=' ')
        {
            c++;
        }
    }
    printf("Total words = %d",c+1);
    return 0;
}
```

56.String Concatenation without Library Function

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int n=0,i;
    gets(a);
    gets(b);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0;b[i]!='\0';i++)
    {
        a[n]=b[i];
        n++;
    }
    a[n]='\0';
    puts(a);
    return 0;
}
```

57.No of Occurring of an alphabet in a string

```
#include<stdio.h>
int main()
{
    char a[100],ch;
    int n=0,i,l=0;
    printf("Enter the string: ");
    gets(a);
    printf("Enter the character: ");
    scanf("%c",&ch);
    while(a[n]!='\0')
    {
        n++;
    }
    for(i=0;i<n;i++)
    {
        if(a[i]==ch)
        {
            l++;
        }
    }
    printf("%d",l);
    return 0;
}
```

58.String Comparison without Library Function

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int n=0,m=0,i,c=0;
    gets(a);
    gets(b);
    while(a[n]!='\0')
    {
        n++;
    }
    while(b[m]!='\0')
    {
        m++;
    }
    if(n!=m)
    {
        printf("String Not Matched");
    }
    else
    {
        for(i=0;i<n;i++)
        {
            if(a[i]==b[i])
            {
                c++;
            }
        }
        if(n==c)
        {
            printf("String Matched");
        }
        else
        {
            printf("String Not Matched");
        }
    }
    return 0;
}
```

59.Matrix Determinant

```
#include<stdio.h>
int main()
{
    int a[3][3],i,j,d;
    printf("Enter The Matrix elements: \n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%d",&a[i][j]);
        }
    }
    d=(a[0][0]*(a[1][1]*a[2][2]-a[1][2]*a[2][1]))-
    (a[0][1]*(a[1][2]*a[2][0]-
a[1][0]*a[2][2]))+(a[0][2]*(a[1][0]*a[2][1]-a[1][1]*a[2][0]));
    printf("The Determinant is %d",d);
    return 0;
}
```

60.Matrix Summation and Subtraction

```
#include<stdio.h>
int main()
{
    int a[3][3],b[3][3],c[3][3],d[3][3],i,j;
    printf("Enter the first matrix element: \n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%d",&a[i][j]);
        }
    }
    printf("Enter the Second matrix element: \n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%d",&b[i][j]);
        }
    }
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            c[i][j]=a[i][j]+b[i][j];
            d[i][j]=a[i][j]-b[i][j];
        }
    }

    printf("Enter the Summation matrix element: \n");
    for(i=0;i<3;i++)
    {
        printf("\n");
        for(j=0;j<3;j++)
        {
            printf ("%d\t",c[i][j]);
        }
    }
}
```

```
printf("Enter the Subtraction matrix element: \n");  
for(i=0;i<3;i++)  
{  
    printf("\n");  
    for(j=0;j<3;j++)  
    {  
        printf ("%d\t",d[i][j]);  
    }  
}  
return 0;  
}
```

61.Binary Search

```
#include<stdio.h>
int main()
{
    int a[100],n,m,first,last,middle,i;
    printf("Enter the number of Array elements: ");
    scanf("%d",&n);
    printf("Enter the Array Elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the Number to be searched: ");
    scanf("%d",&m);
    first=0;
    last=n-1;
    middle=(first+last)/2;
    while(first<=last)
    {
        if(a[middle]<m)
        {
            first=middle+1;
        }
        else if(a[middle]>m)
        {
            last=middle-1;
        }
        else
        {
            printf("The element found in the place
%d",middle+1);
            break;
        }
        middle=(first+last)/2;
    }
    if(first>last)
    {
        printf("The element not found");
    }
    return 0;
}
```

62.Removing All Space from a String

```
#include <string.h>
int main()
{
    char s[1000];
    int i,k=0;
    printf("Enter the string : ");
    gets(s);
    for(i=0;s[i];i++)
    {
        s[i]=s[i+k];
        if(s[i]==' '|| s[i]=='\t')
        {
            k++;
            i--;
        }
    }
    printf("string after removing all blank spaces:\n");
    printf("%s",s);
    return 0;
}
```

63. Lowercase & Uppercase Alphabet

```
#include<stdio.h>
int main()
{
    char ch;
    printf("Enter the character: ");
    scanf("%c",&ch);
    if(ch>='a' && ch<='z')
    {
        printf("lowercase");
    }
    else if(ch>='A' && ch<='Z')
    {
        printf("Uppercase");
    }
    else
    {
        printf("Not a character");
    }
    return 0;
}
```

64.Vowel, Consonant, Number and Special Character Count from a String

```
#include<stdio.h>
int main()
{
    char ch[100],l=0,v=0,c=0,n=0,s=0,i;
    printf("Enter the string: ");
    gets(ch);
    while(ch[l]!='\0')
    {
        l++;
    }
    for(i=0;i<l;i++)
    {
        if(ch[i]=='a' || ch[i]=='e' || ch[i]=='i' || ch[i]=='o' || ch[i]=='u' || c
h[i]=='A' || ch[i]=='E' || ch[i]=='I' || ch[i]=='O' || ch[i]=='U')
        {
            v++;
        }
        else
        if(ch[i]=='1' || ch[i]=='2' || ch[i]=='3' || ch[i]=='4' || ch[i]=='5' || c
h[i]=='6' || ch[i]=='7' || ch[i]=='8' || ch[i]=='9' || ch[i]=='0')
        {
            n++;
        }
        else
        if(ch[i]=='!' || ch[i]=='@' || ch[i]=='#' || ch[i]=='$' || ch[i]=='%' || c
h[i]=='^' || ch[i]=='&' || ch[i]=='*' || ch[i]=='+' || ch[i]=='-
' || ch[i]=='|')
        {
            s++;
        }
        else
        {
            c++;
        }
    }
}
```

```
    }  
}  
printf("No of Vowel %d \n",v);  
printf("No of Consonant %d \n",c);  
printf("No of Number %d \n",n);  
printf("No of Special Character %d \n",s);  
return 0;  
}
```

65.Uppercase to Lowercase & Lowercase to Uppercase

```
#include<stdio.h>
int main()
{
    char ch;
    printf("Enter a Character: ");
    scanf("%c",&ch);
    if(ch>='a' && ch<='z')
    {
        ch=ch-32;
    }
    else if(ch>='A' && ch<='Z')
    {
        ch=ch+32;
    }
    printf("The Converted Character is %c",ch);
    return 0;
}
```

66. Case Conversion in a String

```
#include<stdio.h>
int main()
{
    char a[100];
    int l=0,i;
    printf("Enter the string: ");
    gets(a);
    while(a[l]!=0)
    {
        l++;
    }
    for(i=0;i<l;i++)
    {
        if((a[i]>='a' && a[i]<='z') || (a[i]>='A' && a[i]<='Z'))
        {
            a[i]=CaseConvert(a[i]);
        }
    }
    for(i=0;i<l;i++)
    {
        printf("%c",a[i]);
    }
    return 0;
}

void CaseConvert(char ch)
{
    if(ch>='a' && ch<='z')
    {
        ch=ch-32;
    }
    else if(ch>='A' && ch<='Z')
    {
        ch=ch+32;
    }
    return ch;
}
```

67. Case Conversion in a String Reverse Order

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    int l=0,i,j;
    printf("Enter the string: ");
    gets(a);
    while(a[l]!=0)
    {
        l++;
    }
    for(i=0;i<l;i++)
    {
        if((a[i]>='a' && a[i]<='z') || (a[i]>='A' && a[i]<='Z'))
        {
            a[i]=CaseConvert(a[i]);
        }
    }
    for(i=0,j=l-1;i<l,j>=0;i++,j--)
    {
        b[j]=a[i];
    }
    for(i=0;i<l;i++)
    {
        printf("%c",b[i]);
    }
    return 0;
}

void CaseConvert(char ch)
{
    if(ch>='a' && ch<='z')
    {
        ch=ch-32;
    }
    else if(ch>='A' && ch<='Z')
    {
        ch=ch+32;
    }
    return ch;
}
```

68.String to number Conversion

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    char a[100];
    int d;
    printf("Enter a string:");
    gets(a);
    d=atoi(a);
    printf("%d",d);
    return 0;
}
```

69.IP Address Validation, Classification

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
int main()
{
    char a[100],b[100];
    int l,i,j=0,c[4],d=0,ip[4],e=0;
    printf("Enter The IP Address: ");
    gets(a);
    l=strlen(a);
    for(i=0;i<l;i++)
    {
        if(a[i]!='.')
        {
            b[j++]=a[i];
        }
        if(a[i]=='.' || i==l-1)
        {
            b[j]='\0';
            j=0;
            c[d++]=atoi(b);
        }
    }
    ip[0]=c[0];
    ip[1]=c[1];
    ip[2]=c[2];
    ip[3]=c[3];

    if((ip[0]>=0 && ip[0]<=255) && (ip[1]>=0 && ip[1]<=255) &&
(ip[2]>=0 && ip[2]<=255) && (ip[3]>=0 && ip[3]<=255))
    {
        printf("Valid IP Address.\n\n");
        e++;
    }
    else
    {
        printf("Not Valid IP address");
    }
}
```

```
if(e==1)
{
    if((ip[0]>=0 && ip[0]<=127))
    {
        printf("Class A IP Address");
    }
    else if((ip[0]>=128 && ip[0]<=191))
    {
        printf("Class B IP Address");
    }
    else if((ip[0]>=192 && ip[0]<=223))
    {
        printf("Class C IP Address");
    }
    else if((ip[0]>=224 && ip[0]<=239))
    {
        printf("Class D IP Address");
    }
    else
    {
        printf("Class E IP Address");
    }
}
return 0;
}
```

70. Alphabet frequency in a string

```
#include<stdio.h>
#include<string.h>
int main()
{
    char a[100];
    int l,i,freq[26];
    printf("Enter The String: ");
    gets(a);
    l=strlen(a);
    for(i=0;i<26;i++)
    {
        freq[i]=0;
    }
    for(i=0;i<l;i++)
    {
        if(a[i]>='a' && a[i]<='z')
        {
            freq[a[i]-97]++;
        }
        else if(a[i]>='A' && a[i]<='Z')
        {
            freq[a[i]-65]++;
        }
    }
    printf("The frequency of each alphabet are: \n");
    for(i=0;i<26;i++)
    {
        if(freq[i]!=0)
        {
            printf("%c = %d times\n\n",i+97,freq[i]);
        }
    }
    return 0;
}
```

71.Substring Search in a given String

```
#include<stdio.h>
#include<string.h>
int main()
{
    char a[100],b[100];
    int i,j,m=0,n=0,flag;
    printf("Enter the String: ");
    gets(a);
    printf("Enter The SubString: ");
    gets(b);
    while(a[m]!='\0')
    {
        m++;
    }
    while(b[n]!='\0')
    {
        n++;
    }
    for(i=0;i<=(m-n);i++)
    {
        for(j=i;j<=(i+n);j++)
        {
            flag=1;
            if(a[j]!=b[j-i])
            {
                flag=0;
                break;
            }
        }
        if(flag==1)
            break;
    }
    if(flag==1)
    {
        printf("Found");
    }
    else
    {
        printf("Not Found");
    }
    return 0;
}
```

72.Basic Pointer-01

```
#include<stdio.h>
int main()
{
    int x=10, *px;
    px=&x;
    printf("The Value of X= %d\n",x);
    printf("The Address of X= %d\n",&x);
    printf("The Address of Pointer= %d\n",&px);
    printf("The Value of the Address to be point= %d\n",*px);
    return 0;
}
```

73.Basic Pointer (Summation)

```
#include<stdio.h>
int main()
{
    int x=10,y=20,sum,*p,*p1;
    p=&x;
    p1=&y;
    sum=*p+*p1;
    printf("Sum= %d",sum);
    return 0;
}
```

74.Basic Pointer (Swapping Variable)

```
#include<stdio.h>
int main()
{
    int x=10,y=20,temp;
    int *p1,*p2;
    p1=&x;
    p2=&y;
    temp=*p1;
    *p1=*p2;
    *p2=temp;
    printf("x = %d\n",x);
    printf("y = %d",y);
    return 0;
}
```

75.Basic pointer (Swapping Using Function)

```
#include<stdio.h>
int main()
{
    int x=10,y=20;
    Swapping(&x,&y);
    printf("x = %d\n",x);
    printf("y = %d",y);
    return 0;
}
void Swapping(int *x, int *y)
{
    int temp;
    temp=*x;
    *x=*y;
    *y=temp;
}
```

76.Basic pointer (Array Elements Sum)

```
#include<stdio.h>
int main()
{
    int a[10], i,n,sum=0;
    printf("Enter How many elements in the array : ");
    scanf("%d",&n);
    printf("Enter the elements ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {
        sum=sum+*(a+i);
    }
    printf(" Sum is =%d",sum);
    return 0;
}
```

77.String Reverse Using Pointer

```
#include<stdio.h>
int main()
{
    char a[100];
    int i,l;
    printf("Enter a String: ");
    gets(a);
    l=strlen(a);
    for(i=l-1;i>=0;i--)
    {
        printf("%c",*(a+i));
    }
    return 0;
}
```

78.String Concatenation Using Pointer

```
#include<stdio.h>
int main()
{
    char a[100],b[100];
    printf("Enter The First String : ");
    gets(a);
    printf("Enter The Second String : ");
    gets(b);
    char *aa=a;
    char *bb=b;
    while(*aa)
    {
        aa++;
    }
    while(*bb)
    {
        *aa=*bb;
        bb++;
        aa++;
    }
    *aa='\0';
    printf("%s",a);
    return 0;
}
```

79.Mountain sequence or not

Input:

6

1 2 1 1 3 4

Output: Not Mountain

Input:

6

1 2 1 3 2 4

Output: Mountain

Idea: To check if the sequence is mountain we have to make sure that no neighbouring elements are the same.

```
#include<stdio.h>
int main()
{
    int a[10],n,i,c=0;
    printf("Enter how many elements: ");
    scanf("%d",&n);
    printf("Enter the Elements: ");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {
        if(a[i]==a[i+1])
        {
            c++;
            break;
        }
    }
    if(c==1)
    {
        printf("Not Mountain Sequence");
    }
    else
    {
        printf("Mountain Sequence");
    }
    return 0;
}
```